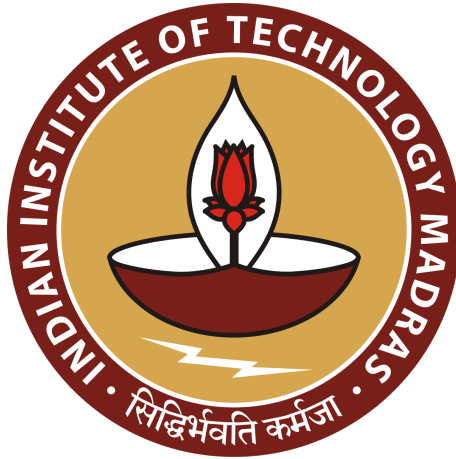


Department Of Aerospace Engineering,
Indian Institute Of Technology Madras



**Control Algorithm Implementation to
match thrust of two coaxial rotors**

Contents

1	Problem Statement	3
2	Theory of thrust in coaxial rotor configuration	3
2.1	Momentum Theory and BET Analysis	3
2.2	BEMT Analysis - With tip loss factor	3
3	Hardware operating concepts	4
3.1	Motors:	4
3.2	Load Cells:	4
3.3	Electronic speed controllers (ESC):	4
4	Experimental Set up	5
4.1	Hardware Set up	5
4.1.1	Experimental setup	5
4.1.2	Connections	5
4.2	Electrical Schematic	6
5	Open Loop Results	7
6	Closed Loop Control	10
6.1	Block Diagram	10
6.2	Controller Flow	10
6.2.1	Single Rotor - P controller	10
6.2.2	Coaxial Setup - P Controller	10
6.3	Results	11
6.3.1	h=15cm	11
6.3.2	h=22.4cm	12
6.3.3	h=30cm	14
6.3.4	Effect of the coaxial rotors	15
6.3.5	Comparison with initial momentum theory and BEMT results . .	17
7	Optimally Powered Thrust Generation	17
8	Appendix	17
8.1	Momentum Theory Analysis Code	17
8.2	BEMT + Tiploss Analysis Code	18

List of Figures

1	Coaxial System	3
2	ESC	5
3	Experimental setup	5
4	Circuit Schematic	6
5	Open Loop Block Diagram	7
6	PWM To RPM Mapping	7
7	Control Schematic Block Diagram	10
8	Thrust vs time for total 100g target thrust - 15cm	11

9	Thrust vs time for total 150g target thrust - 15cm	12
10	Thrust vs time for total 200g target thrust - 15cm	12
11	Thrust vs time for total 100g target thrust - 22.4cm	13
12	Thrust vs time for total 150g target thrust - 22.4cm	13
13	Thrust vs time for total 200g target thrust - 22.4cm	14
14	Thrust vs time for total 100g target thrust - 30cm	14
15	Thrust vs time for total 150g target thrust - 30cm	15
16	Thrust vs time for total 200g target thrust - 30cm	15
17	Thrust vs PWM for the lower rotor as time passes	16
18	Thrust vs PWM for the upper rotor as time passes	16

1 Problem Statement

Implement a hardware based control algorithm to match the thrust of two rotors in a coaxial rotor configuration.

2 Theory of thrust in coaxial rotor configuration

In simple terms, two or more rotors that have the same axis of rotation whose rotor planes are kept at some finite distance from each is called a Coaxial Rotor System. The main advantage behind using such a system stems from the fact that we can spin the two rotors in opposite directions to do away with a tail rotor in helicopters (For Yaw Trim)



Figure 1: Coaxial System

2.1 Momentum Theory and BET Analysis

Utilising Momentum Theory and Blade Element Theory and assuming uniform inflow, we can derive an implicit equation for CT and induced inflow. Using this analysis, we compute the effective K value used in the thrust expression and it is seen that the effective K value is obtained as $CT_{lower} = 0.68CT_{upper}$

2.2 BEMT Analysis - With tip loss factor

Similarly, utilising BEMT and including a tip loss factor for more realistic results, another analysis was carried out. The inflow from the top rotor was assumed to be concentrated into half the area of the bottom rotor as per momentum theory results and thereby the remaining area of the bottom rotor was assumed to have no effect from the inflow of the top rotor.

Doing so gave a $CT_{lower} = 0.7CT_{upper}$

3 Hardware operating concepts

3.1 Motors:

The motor used for each of the rotors is a 1400KV BLDC motor, or brushless DC motor commonly used for quadcopters and multirotors. A 30A 2-4S ESC is used to control the voltage PWM input to the motor.

ESC uses a PWM signal to manipulate the speed of the motor through variations in the duty cycle of the PWM signal. The motor comprises a stator with coils and a rotor with magnets. The ESC adjusts the duration of power pulses to the stator coils using PWM, influencing the strength of the electromagnetic field. This causes the rotor, having permanent magnets, to turn at varying speeds according to the strength of the magnetic field.

3.2 Load Cells:

The load cell used to measure rotor thrusts is the **HX711** load cell. Two load cells are mounted each at the base of one of the two coaxial rotors.

A load cell is a transducer that converts force or load into an electrical signal. It typically consists of a metal structure with strain gauges attached.

When a force is applied to the load cell, it deforms slightly, causing strain in the attached gauges. This strain results in a change in electrical resistance in the gauges. The load cell then measures this change and converts it into an electrical signal, usually in the form of voltage. The magnitude of the voltage correlates with the applied force or weight.

3.3 Electronic speed controllers (ESC):

Electronic speed controllers are responsible for controlling speed and direction of the motors by regulating the current supplied to them. The ESC's used in the experiment were rated 30A 2-4S. These controllers translate input signals into precise output power levels, dictating the motor's RPM and direction. ESC's utilize PWM techniques to regulate the power supplied to the motor. By adjusting the duty cycle of the PWM signal, they control the average voltage and current sent to the motor, thereby controlling its speed.

ESC's have inbuilt PWM circuitry, MOSFETS and come with a BEC (Battery eliminator circuit)



Figure 2: ESC

4 Experimental Set up

4.1 Hardware Set up

4.1.1 Experimental setup

Two propellers, one attached to each BLDC motor are kept facing each other with both their wake's in downward direction. Motor situated at the top is mounted to a surface which is attached firmly to a steel rod whose height is adjustable, thereby allowing to perform the experiment with varying separation between both the motors.



Figure 3: Experimental setup

4.1.2 Connections

- VCC of Load cell amplifiers to 5V of Arduino

- GND of Load cell amplifiers to GND
- Data line of Load cell 1 to digital pin 2 (PE4)
- Data line of Load cell 2 to digital pin 4 (PG5)
- Serial Clock of Load cell 1 to digital pin 3 (PE5)
- Serial Clock of Load cell 1 to digital pin 5 (PE3)
- PWM of ESC 1 to digital pin 8 (PH5)
- PWM of ESC 2 to digital pin 9 (PH6)
- ESC's to motors
- ESC's to power supply

4.2 Electrical Schematic

Shown below is the schematic of the circuit, components include Arduino Mega 2560, Load cell amplifiers HX711, 30A 2-4S ESC's and 1400 KV rated BLDC motors.

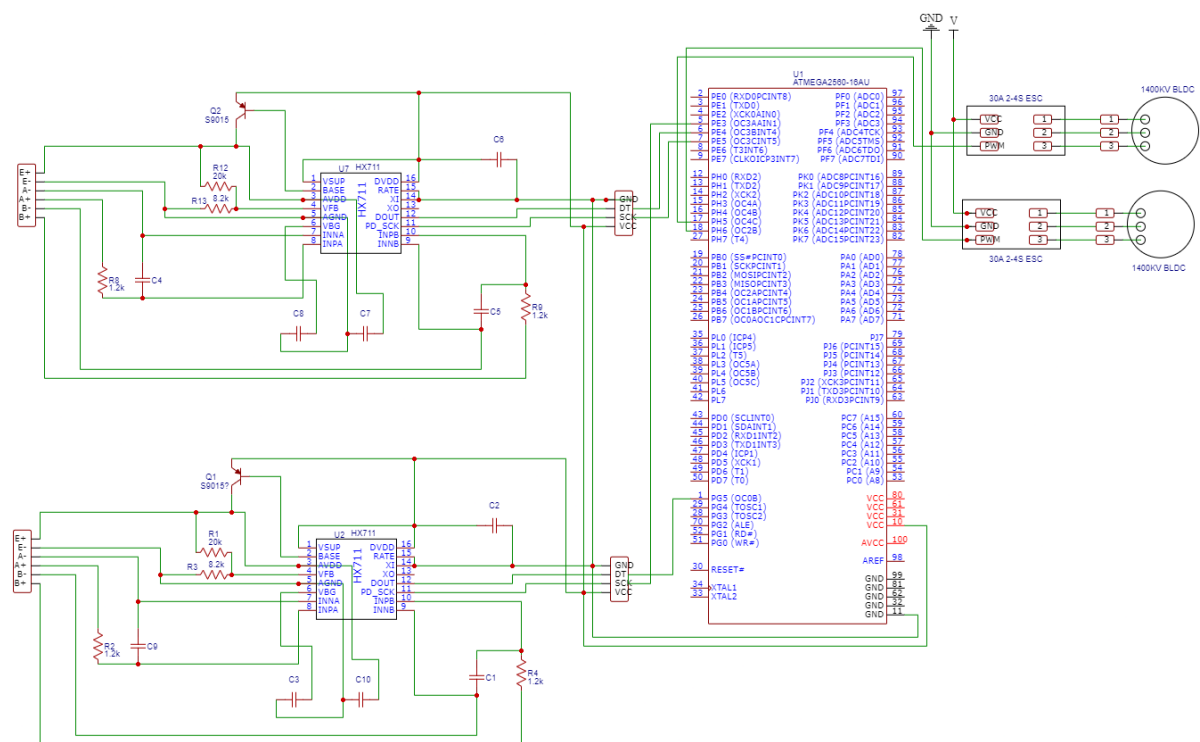


Figure 4: Circuit Schematic

5 Open Loop Results

A simple block diagram of an open loop set up is as shown:

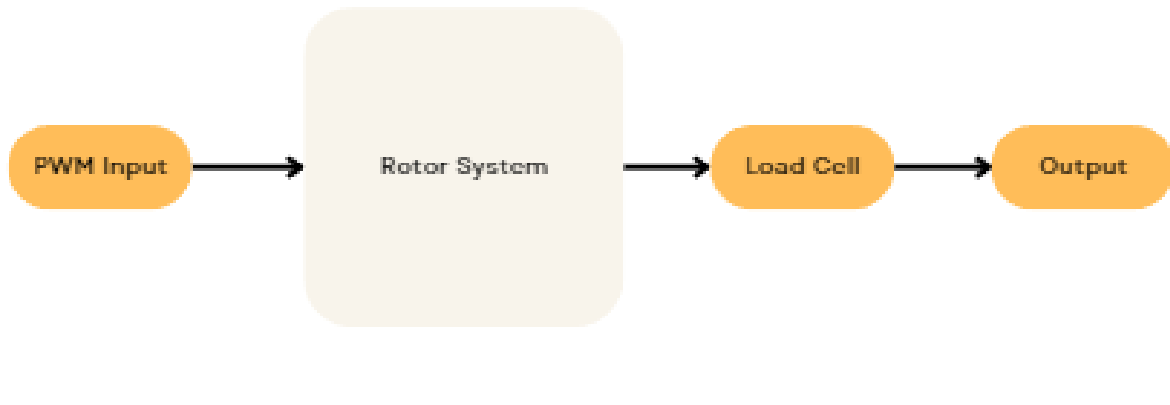


Figure 5: Open Loop Block Diagram

By sweeping across a range of PWM inputs from 1050 to 2000 (in steps of 50). We get the following mapping.

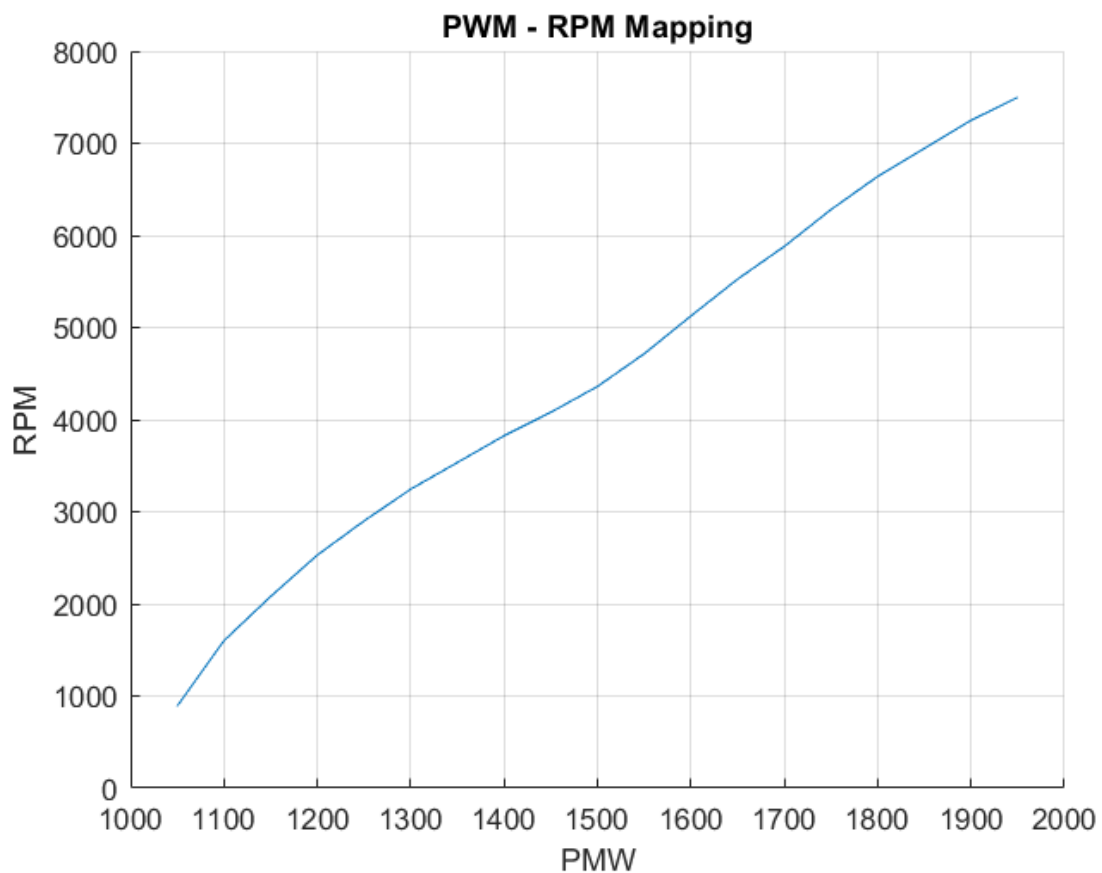
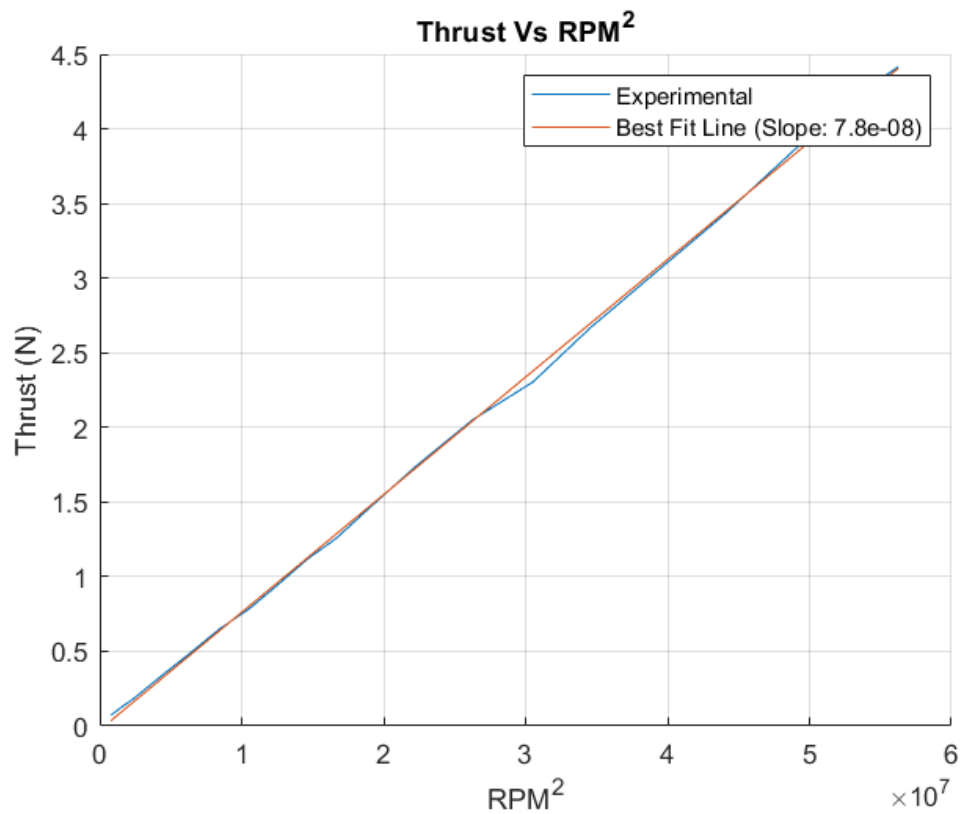
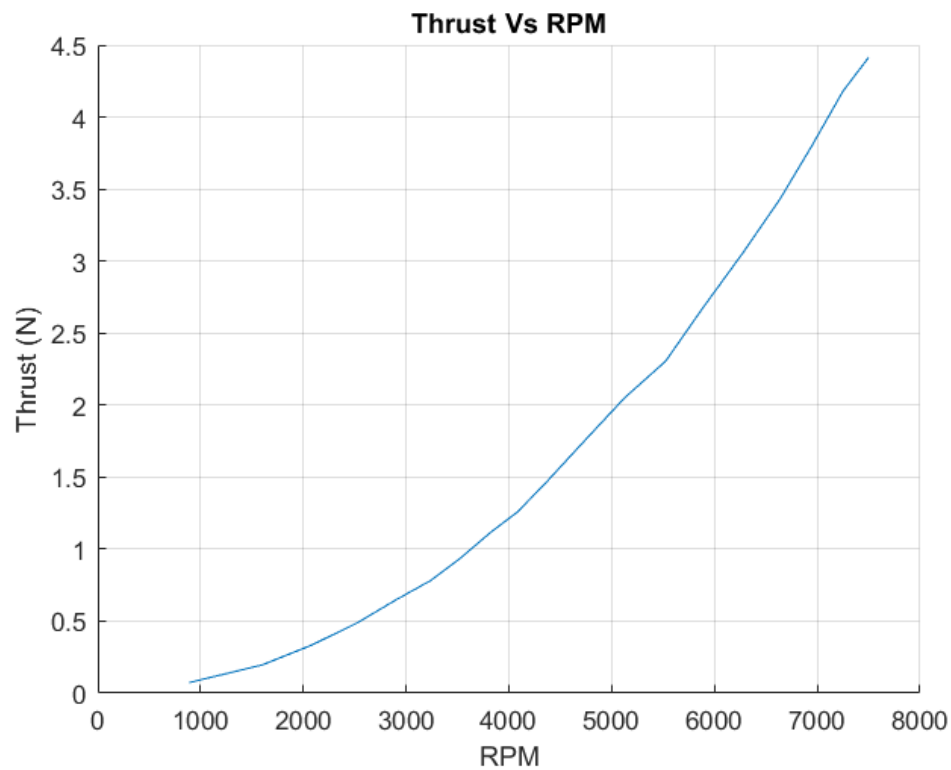
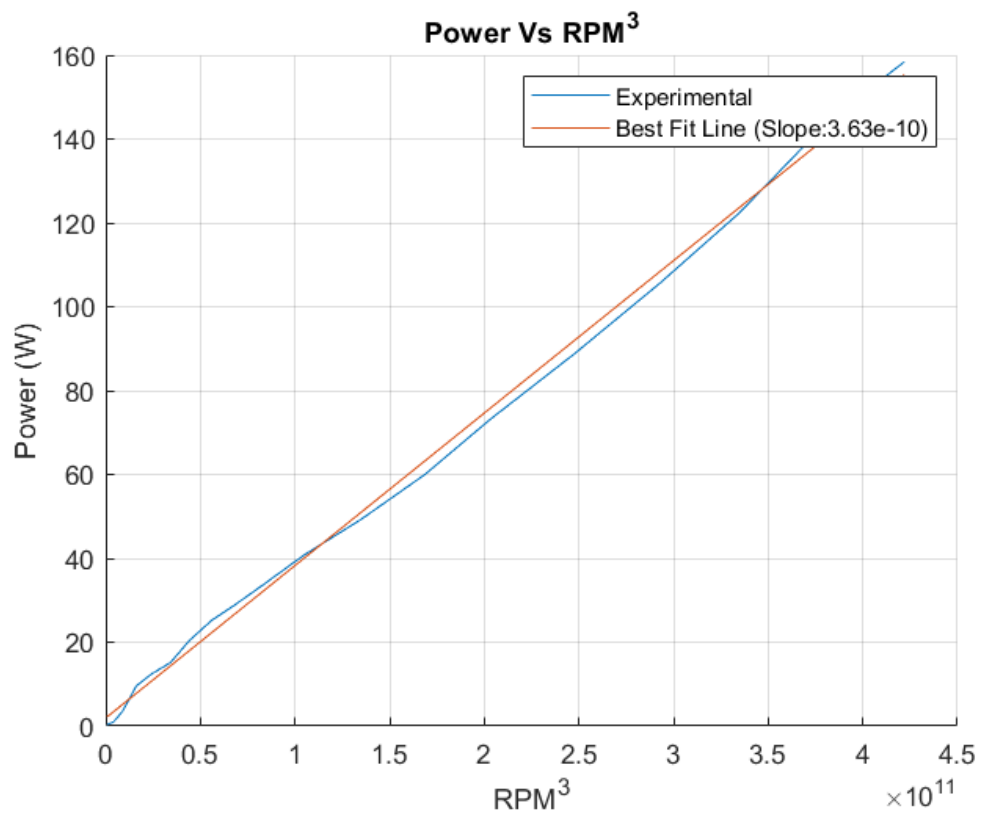
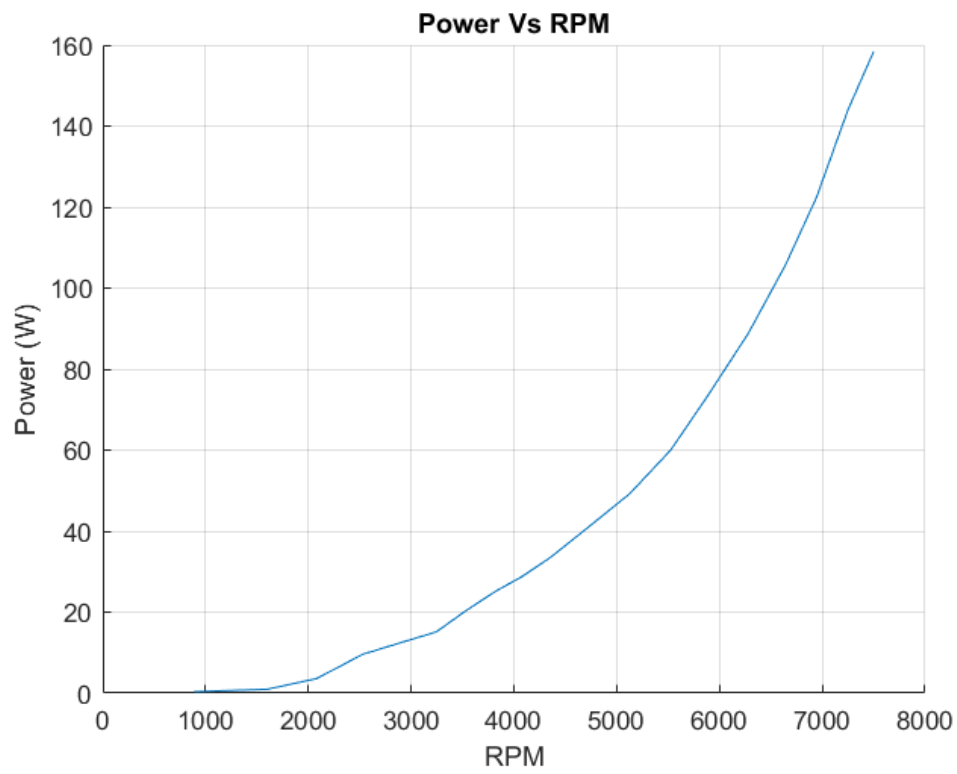


Figure 6: PWM To RPM Mapping

We also plot the thrust produced corresponding to each RPM. We find that there is a linear relationship between thrust and the square of RPM. (Which is expected and is in agreement with theoretical concepts as discussed above)



Similarly we also plot the results of Power Vs RPM. We observe here that there is a linear relation between the cube of RPM with Power. (Once again agreeing with theory).



6 Closed Loop Control

6.1 Block Diagram

:

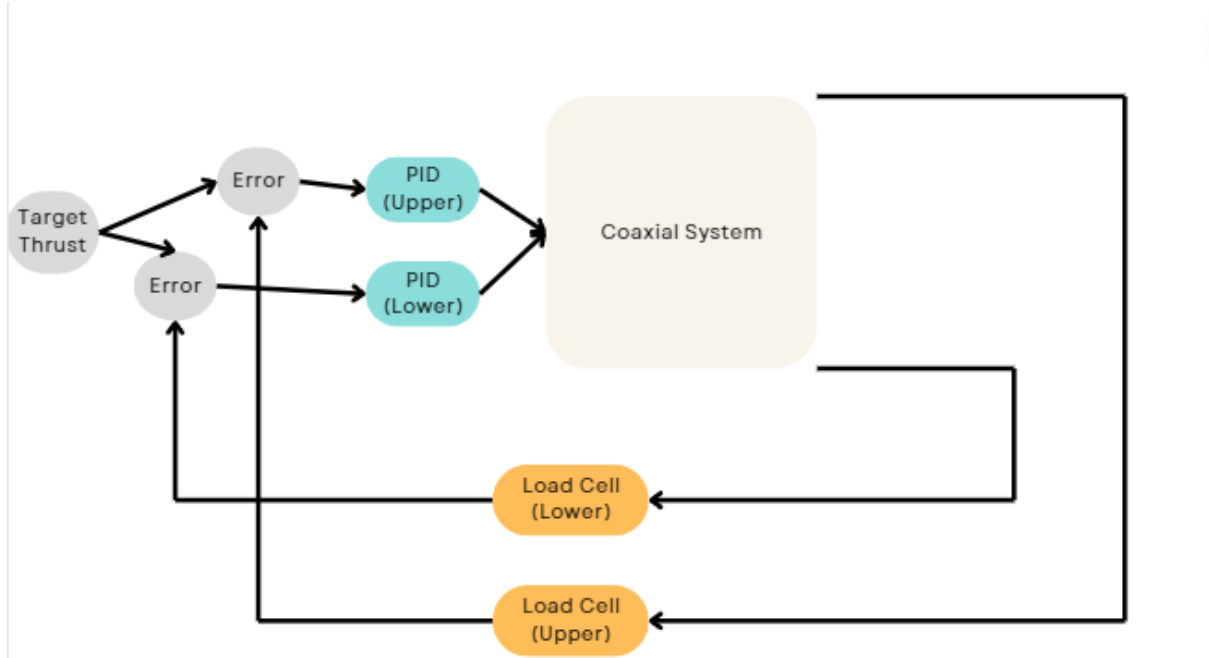


Figure 7: Control Schematic Block Diagram

The control architecture was made as shown in the block diagram. Due to there being no necessity to couple the control inputs for the upper and lower rotors, we created two parallelly running PID loops to attain the matched thrust.

6.2 Controller Flow

6.2.1 Single Rotor - P controller

Initially, we utilised a proportional only controller to control a single rotor (because of the way our architecture is defined, we can just duplicate this setup). We then noticed that a simple P controller is also sufficient and will not lead to steady state error due to the lack of dynamics in the system. Suppose your target thrust is 400g, as long as you are not at 400g, the error term is non zero and the control inputs are then changed keeping in mind the previous instant's value thereby always helping attain the required thrust.

6.2.2 Coaxial Setup - P Controller

As mentioned, a P controller is sufficient for the setup. However, due to the overshoots that we saw, we wanted to utilise a predictive aspect to help settle down closer to the target value faster. Therefore, we added a Derivative term as well.

6.3 Results

As mentioned, we finally used a PD controller to obtain our targets. 3 different heights, and 3 different target thrusts for each height were considered.

6.3.1 $h=15\text{cm}$

1. Total target thrust = 100g of force The target thrust for each rotor was set as 50g.

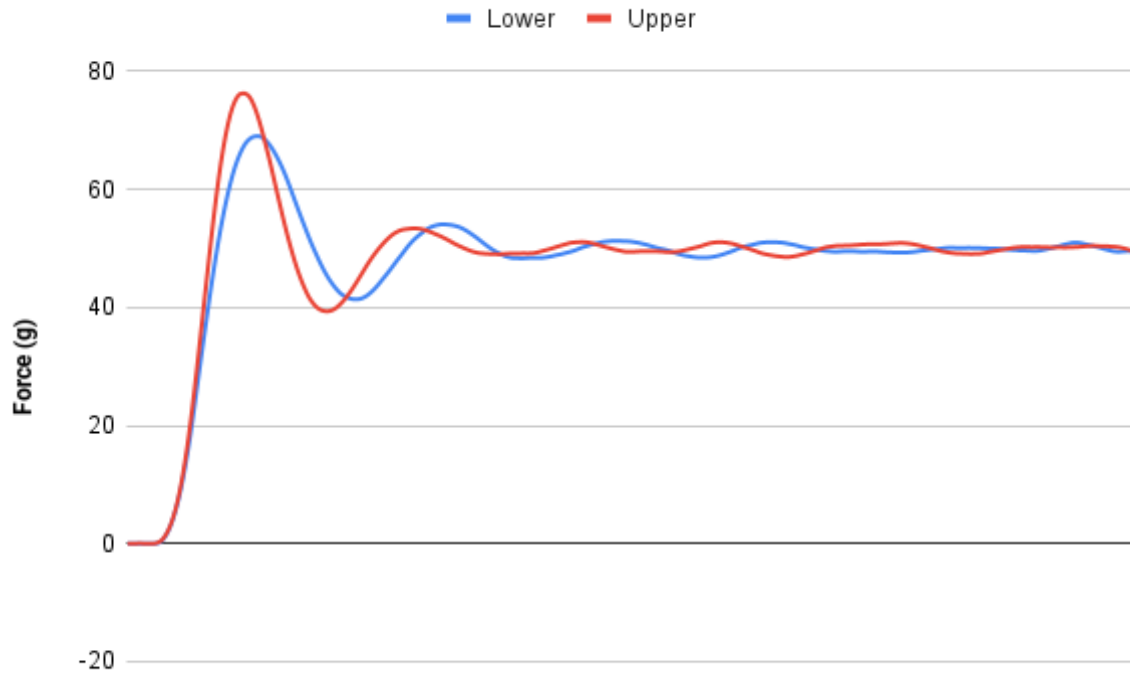


Figure 8: Thrust vs time for total 100g target thrust - 15cm

2. Total target thrust = 150g of force Target thrust for each rotor was set as 75g.

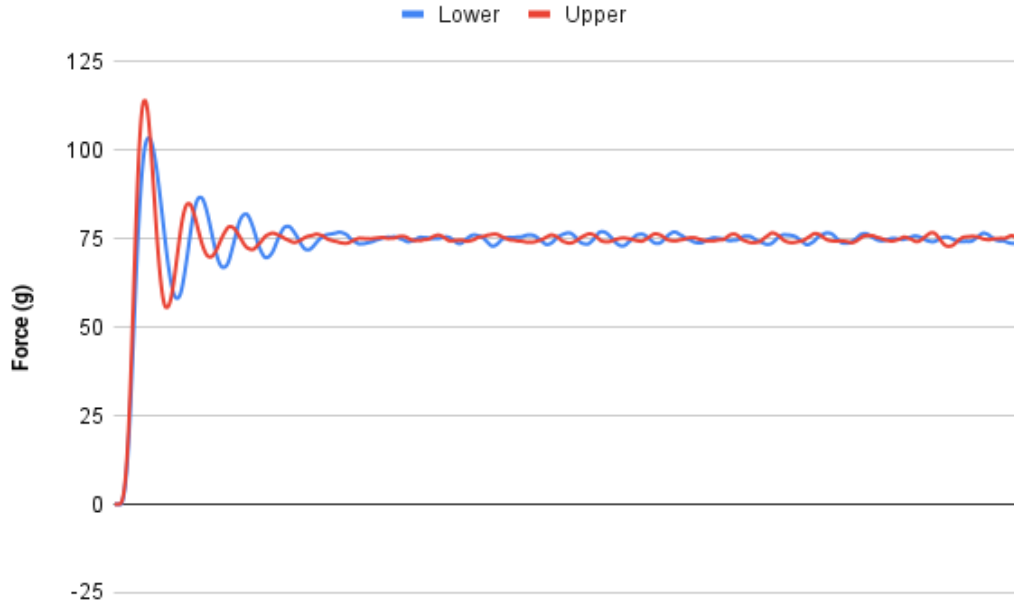


Figure 9: Thrust vs time for total 150g target thrust - 15cm

3. Total target thrust = 200g of force Target thrust for each rotor was set as 100g.

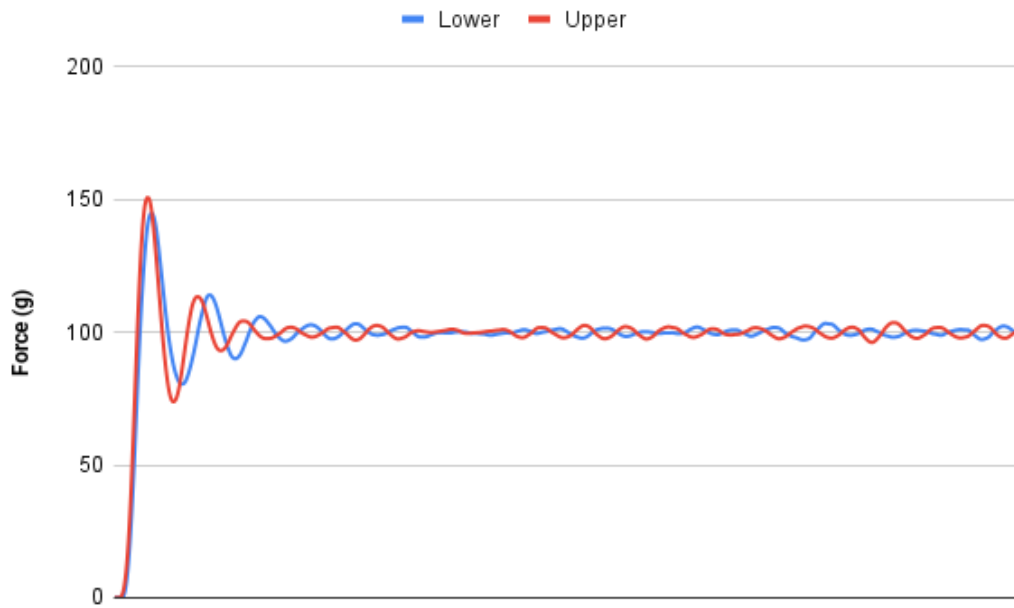


Figure 10: Thrust vs time for total 200g target thrust - 15cm

6.3.2 h=22.4cm

1. Total target thrust = 100g of force The target thrust for each rotor was set as 50g.

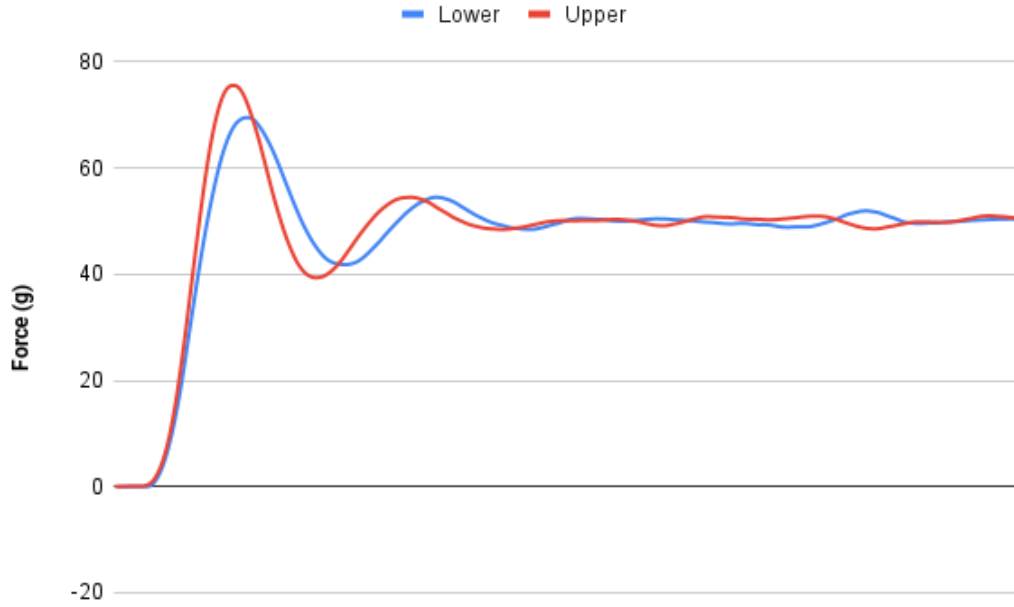


Figure 11: Thrust vs time for total 100g target thrust - 22.4cm

2. Total target thrust = 150g of force Target thrust for each rotor was set as 75g.

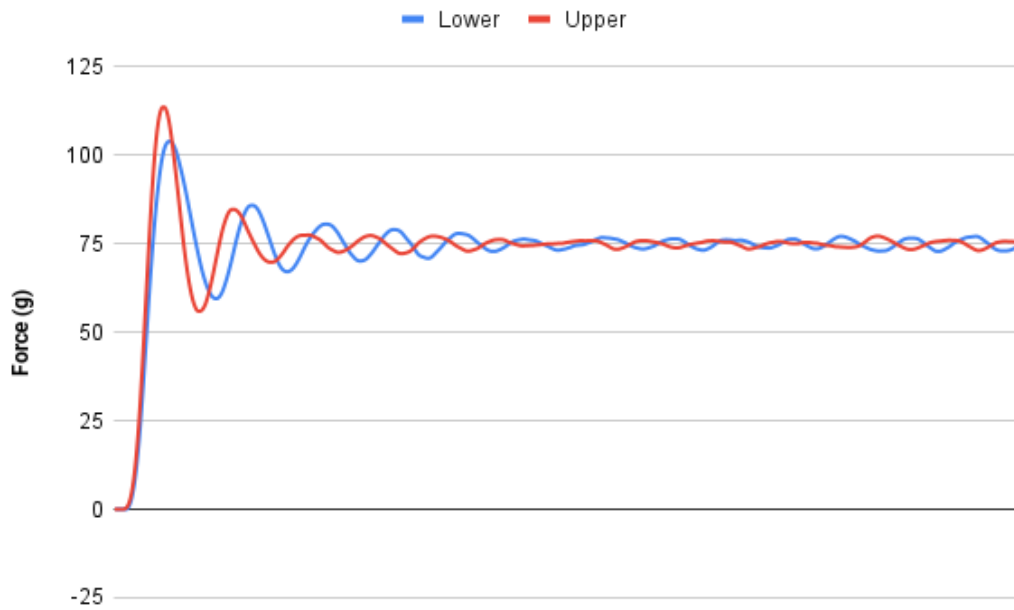


Figure 12: Thrust vs time for total 150g target thrust - 22.4cm

3. Total target thrust = 200g of force Target thrust for each rotor was set as 100g.

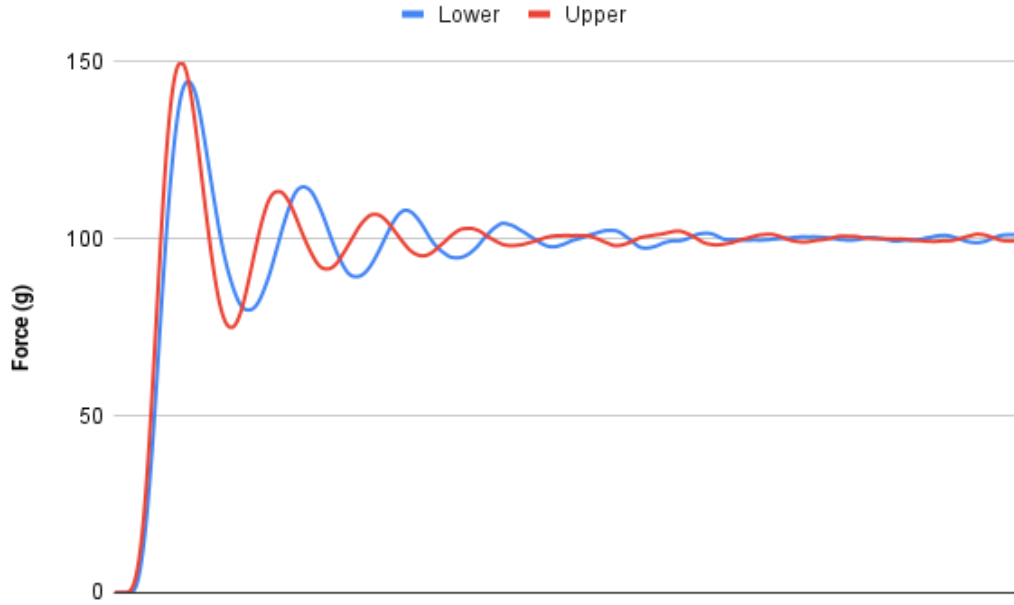


Figure 13: Thrust vs time for total 200g target thrust - 22.4cm

6.3.3 h=30cm

1. Total target thrust = 100g of force The target thrust for each rotor was set as 50g.

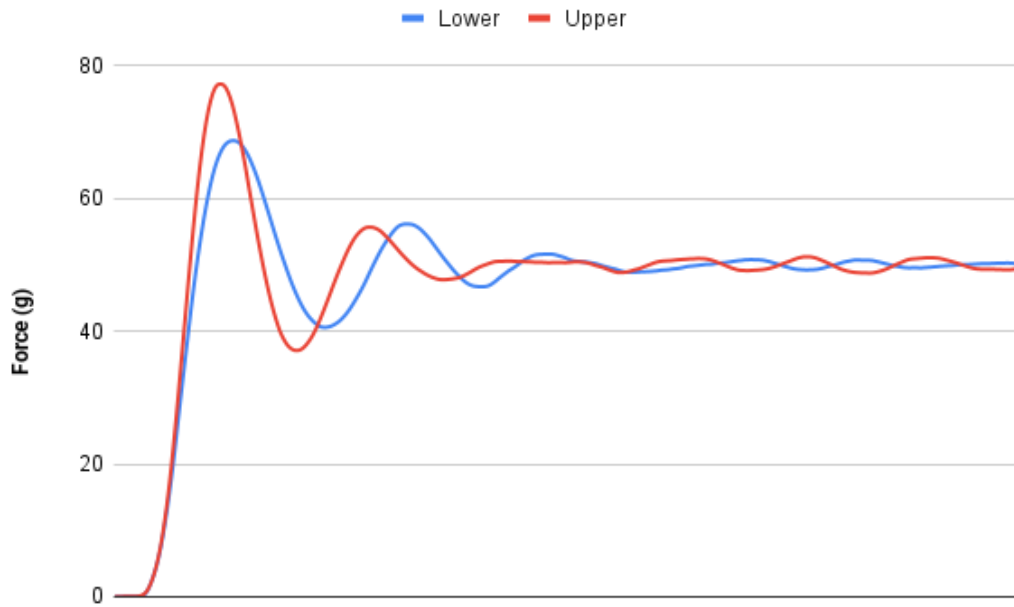


Figure 14: Thrust vs time for total 100g target thrust - 30cm

2. Total target thrust = 150g of force Target thrust for each rotor was set as 75g.

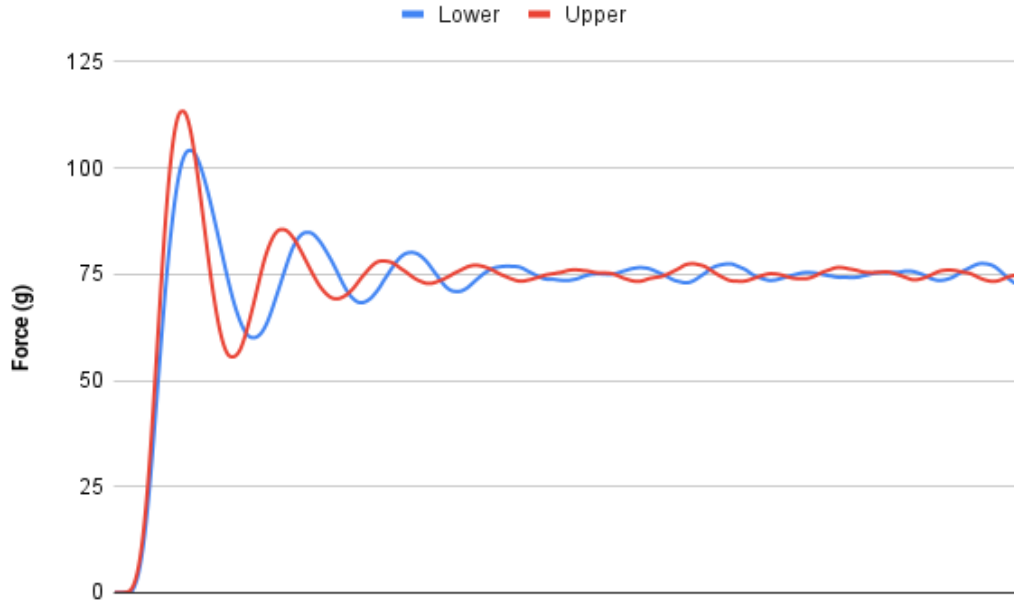


Figure 15: Thrust vs time for total 150g target thrust - 30cm

3. Total target thrust = 200g of force Target thrust for each rotor was set as 100g.

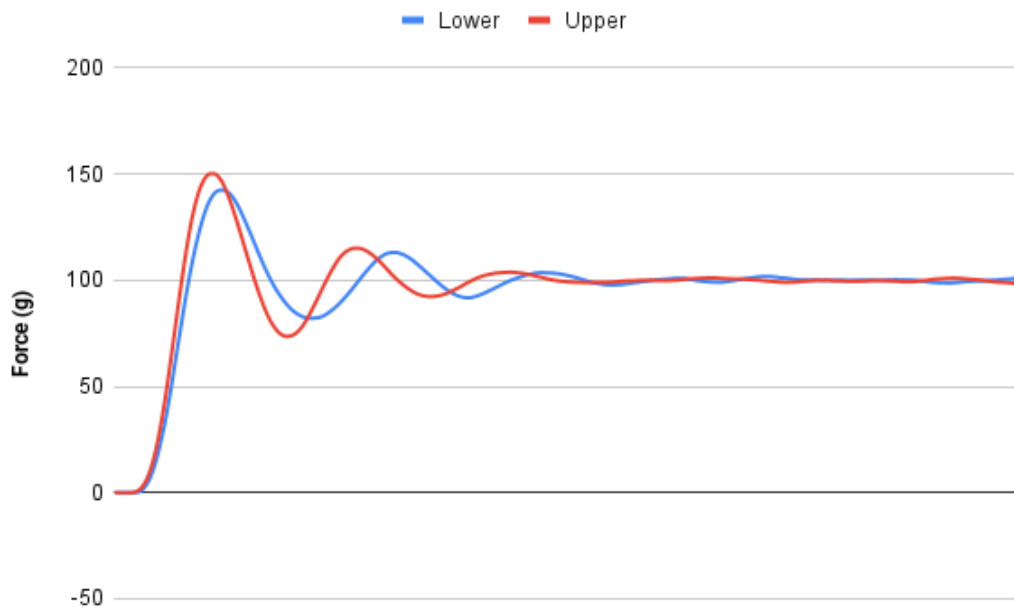


Figure 16: Thrust vs time for total 200g target thrust - 30cm

As seen from all the results, we are achieving the target setpoint. There are some discrepancies due to delays in the load cell reading and possibly some wind disturbances as well but we see overall satisfactory control about the setpoint.

6.3.4 Effect of the coaxial rotors

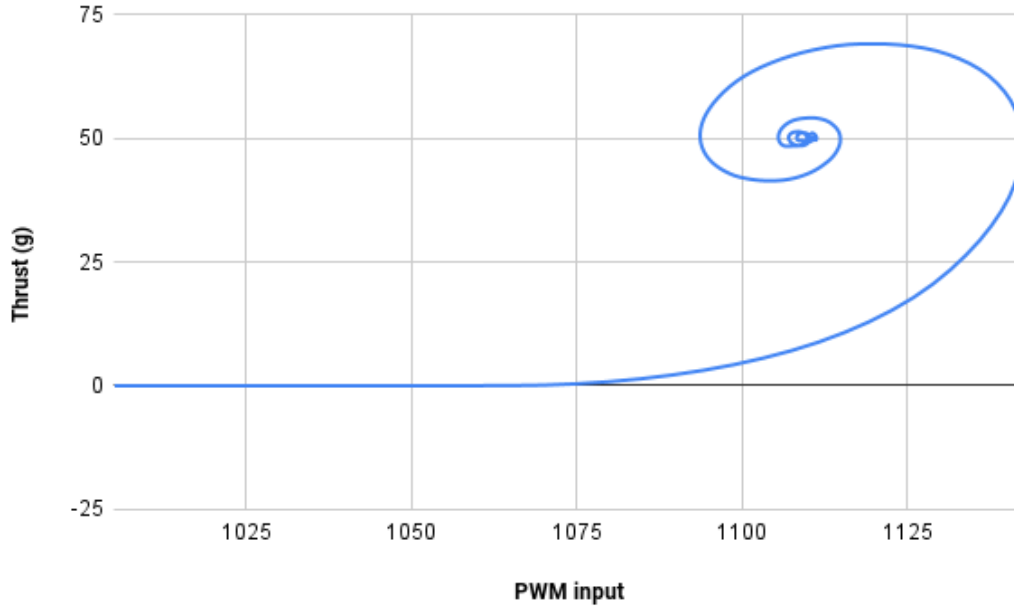


Figure 17: Thrust vs PWM for the lower rotor as time passes

Initially assumed that this spiralling nature was due to the wake of the top rotor felt at the height of the bottom rotor, the same figure for the top rotor was also plotted and was seen to be as shown below.

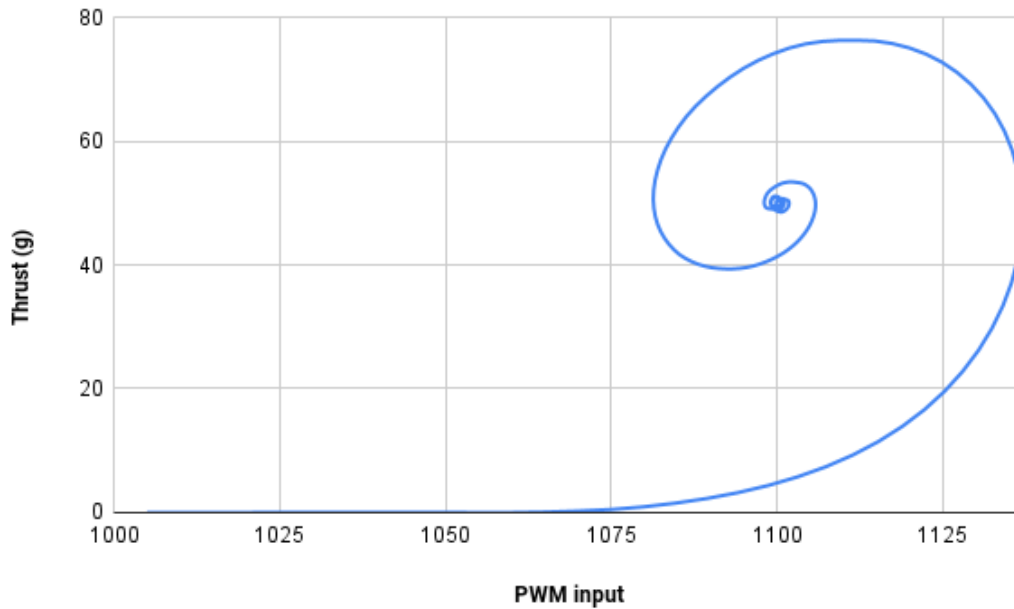


Figure 18: Thrust vs PWM for the upper rotor as time passes

As a similar spiral is also seen here, we can deduce that the spiral is not (atleast not purely) due to the effect of the wake at the bottom but rather, due to dynamic inflow and delays in the system with the sensors and the code.

6.3.5 Comparison with initial momentum theory and BEMT results

From the experimental data, we can obtain the effective K value which is seen to be around 0.8 from different test cases.

The theoretical analysis seems to give us a value from momentum theory of 0.68. However, utilising the fact that only half the area of the bottom rotor has the effect of the top rotor felt on it, we obtain a value of approximately 0.8.

However, the BEMT analysis gives us the result as 0.7.

7 Optimally Powered Thrust Generation

If we require a certain total thrust force to be generated and we want to do it in an optimal fashion as to draw the least power overall, a simple code can be written to achieve this result.

```
clear ;
clc ;
close all ;

fmincon(@(rpm) costfn(rpm),100*ones(2,1),[],[],[],[],[],[],[],...
    @(rpm) nonlincon(rpm))

function cost = costfn(rpm)
    K = 0.8;
    % rpm_u = rpm(1);
    % rpm_l = rpm(2);

    cost = rpm(1)^3+K^1.5*rpm(2)^3
end

function [c,ceq] = nonlincon(rpm)
    K = 0.8;
    CT = 1e-4;
    c = -rpm;
    ceq = CT*(rpm(1)^2+K*rpm(2)^2) - 50*10^-3;
end
```

This can be implemented on the hardware by mapping back the RPM vs PWM graph and thereafter commanding the required PWM from each rotor.

8 Appendix

8.1 Momentum Theory Analysis Code

```
clear ; clc ; close all ;
sigma=0.05;
cla=2*pi ;
thetavec=0.01:0.01:0.2;
N=length(thetavec);
```

```

CTu=zeros(N,1);
CTL=zeros(N,1);
for i=1:N
    theta=thetavec(i);
    p=[1 sigma/8*cla -sigma/12*theta*cla];
    lambdai=max(roots(p));
    pL=[1 sigma/8*cla+lambdai sigma/8*cla*lambdai-1/12*sigma*cla*theta];
    lambdaL=max(roots(pL));
    CTu(i)=2*lambdai^2;
    CTL(i)=2*(lambdaL+lambdai)*lambdaL;
end
pfit=polyfit(CTu,CTL,1);
plot(CTu,CTL,'*r',CTu,polyval(pfit,CTu),'-b');

```

8.2 BEMT + Tiploss Analysis Code

```

clear;clc;close all;
sigma=0.05;
cla=2*pi;
thetavec=0.01:0.01:0.2;
N=length(thetavec);
CTu=zeros(N,1);
CTL=zeros(N,1);
for i=1:N
    theta=thetavec(i);

    r = 0.01:0.01:1;
    x0 = 0.8*ones(2,length(r));
    Nb = 3;
    soln = fsolve(@(x) tiploss(x, Nb, sigma, cla, theta, r, 0), x0);
    CTu(i) = 0.01*trapz(4*soln(2,:).*soln(1,:).^2.*r);

    r_mod = linspace(0.01,1,71);
    pfit = polyfit(r,soln(1,:),1);

    lambda_felt = polyval(pfit,r_mod);
    lambda_felt(end+1:100) = 0;
    soln2 = fsolve(@(x) tiploss(x, Nb, sigma, cla, theta, r,...
    lambda_felt), x0);
    CTL(i) = 0.01*trapz(4*soln2(2,:).*soln2(1,:).*...
    (soln2(1,:)+soln(1,:)).*r);
end
pfit=polyfit(CTu,CTL,1);
plot(CTu,CTL,'*r',CTu,polyval(pfit,CTu),'-b');

1/pfit(1)

function temp = tiploss(x, Nb, sigma, C_lalp, theta, r, lambda_c)

```

```

lambda = x(1,:);
F = x(2,:);

temp(1,:) = F.*8.*lambda.*(lambda+lambda_c) - ...
sigma.*C_lalp.*(theta-(lambda+lambda_c)./r).*r;
temp(2,:) = F - 2/pi.*acos(exp(-(Nb.*(1-r)./(2.*lambda))));
end

```